

CryorApply - System Design Document

AI-powered CV reviewer for students and beginner job seekers.

1. Introduction

CryorApply is a web application that helps students and beginner job seekers improve their CVs before applying for jobs or internships. The platform accepts a CV file in PDF or DOCX format, extracts the text, analyses the content with an AI service, and returns structured feedback with an overall score and practical improvement suggestions.

The current implementation plan uses a React/Vite frontend, a FastAPI backend, Supabase for authentication, database, and storage, and an AI provider for CV analysis.

2. Team and Roles

Name	Role	Responsibilities
Alexandr An	Team Lead / Backend Developer	Team coordination, task tracking, GitHub updates, weekly progress notes, backend API development, CV upload processing, and backend integration support
Konstantin Shevtsov	AI & Testing Specialist / Frontend Developer	AI prompt design, CV review testing, sample CV preparation, feedback quality improvements, frontend structure, user flow, and results page integration
Mikhail Makarchuk	Frontend Developer / UI/UX Designer	UI design, CV upload page, authentication screens, result page layout, and user experience improvements
Maksim Pinchuk	Backend Developer / Data Integration	PDF/DOCX text extraction, Supabase database and storage, authentication backend support, and shared backend implementation tasks with Alexandr An

3. System Architecture Overview

The system is a three-tier web application:

- 1 Client layer: React/Vite frontend.
- 2 API layer: FastAPI backend.
- 3 External services: Supabase and an AI analysis provider.

The frontend communicates with the backend through REST API requests. The backend validates uploaded files, extracts CV text, sends the text to the AI analysis service, formats the response, and stores user and review data in Supabase.

3.1 Client Layer

The client layer is developed with React and Vite. It contains the main user-facing pages and components:

- CV upload page: drag-and-drop or file picker interface, file type validation, and upload status.
- Results page: overall CV score, feedback categories, improvement tips, and previous review summary.
- Authentication pages: sign-up and sign-in screens connected to Supabase Auth.
- Shared UI components: buttons, upload panel, feedback cards, score gauge, and navigation.

Main responsibility: Konstantin Shevtsov and Mikhail Makarchuk.

3.2 API Layer

The backend layer is developed with FastAPI. It acts as the main entry point for frontend requests and controls the CV review workflow.

Current and planned backend responsibilities:

- Health check endpoint for deployment and testing.
- CV upload endpoint that accepts PDF and DOCX files.
- File validation for file type and file size.
- Text extraction from PDF and DOCX files.
- AI service orchestration.
- Response formatting and validation.
- Supabase authentication, database, and storage integration.

Main responsibility: Alexandr An and Maksim Pinchuk.

3.3 AI Review Layer

The AI review layer receives extracted CV text and returns structured JSON feedback. The prompt asks the AI service to evaluate the CV across several categories:

- Structure
- Skills
- Experience
- Formatting
- Role match

The AI output must include an overall score, a short summary, category-level feedback, and practical next steps. The backend validates the response before returning it to the frontend.

Main responsibility: Konstantin Shevtsov, with backend integration support from Alexandr An.

3.4 External Services

Supabase Auth

Supabase Auth manages user accounts and JWT-based authentication. The frontend uses it for sign-up and sign-in, while the backend verifies user identity before storing or returning user-specific review data.

Supabase PostgreSQL

Supabase PostgreSQL stores user-related data, CV upload records, and AI review results.

Supabase Storage

Supabase Storage stores original uploaded CV files in a secure bucket. Each stored file is linked to a user and review record.

AI Provider

The AI provider analyses extracted CV text and returns structured feedback. The provider can be changed later as long as the backend keeps the same response schema for the frontend.

4. Request Workflow

Step	Description	Actor
1. Upload CV	User selects or drops a PDF/DOCX file on the upload page. The frontend validates basic file information and sends the file to the backend.	User / Frontend
2. Validate file	Backend checks file type, size, and request format. Invalid files return a clear error response.	FastAPI Backend
3. Extract text	File processor extracts readable text from PDF or DOCX content.	Backend File Processor
4. AI review	Backend builds the structured prompt and sends extracted text to the AI provider.	AI Review Layer
5. Format response	Backend validates and normalises the AI response into a fixed JSON schema.	Backend
6. Save data	Backend stores the CV file and review result in Supabase.	Backend / Supabase
7. Show results	Frontend displays score, feedback cards, and improvement tips.	Results Page

5. Data Model

Core tables planned in Supabase PostgreSQL:

Table	Key Columns	Description
users	id, email, created_at	Supabase Auth users
cv_uploads	id, user_id, file_path, uploaded_at	Uploaded CV file references

Table	Key Columns	Description
cv_reviews	id, cv_id, score, feedback_json, created_at	AI review results

6. Technology Stack

Layer	Technology	Purpose
Frontend	React, Vite, CSS	User interface and local development server
Backend	Python, FastAPI, Uvicorn	REST API and business logic
AI service	AI provider API	CV analysis and structured feedback
Database	Supabase PostgreSQL	User data and review results
File storage	Supabase Storage	CV file storage
Auth	Supabase Auth	User authentication
Deployment	Vercel / Docker / cloud hosting	Hosting and deployment

7. Key API Endpoints

Method	Endpoint	Description
GET	/health	Check that the backend API is running
POST	/api/cv/review	Upload CV file and receive structured review feedback
POST	/api/auth/signup	Register a new user
POST	/api/auth/login	Sign in and return an auth token
GET	/api/cv/reviews	List a user's past reviews
GET	/api/cv/review/{id}	Get one saved review result

8. Current Prototype Status

The repository currently includes:

- React/Vite frontend prototype with upload flow and sample results.
- FastAPI backend skeleton with health check and CV review endpoint.
- Mock CV analysis response for early frontend and backend integration.
- API contract documentation.
- AI prompt design documentation.

The next development step is to replace mock text extraction and mock AI feedback with real PDF/DOCX parsing and AI service integration.